Formulae & Definitions — Topic 13 · Gravitational Fields

Newton's law of gravitation 万有引力定律 — the force between two point masses is proportional to the product of their masses and inversely proportional to the square of their separation: $F = GMm/r^2$.

Universal gravitational constant — $G = 6.67 \times 10^{-11}$ N m² kg⁻².

Uniform sphere — the whole mass may be taken as a point mass at its centre (r measured from the centre).

Gravitational field 引力场 — a region of space where a mass experiences a force; field lines show the direction of the force (the closer the lines, the stronger the field).

Field strength 场强 g — the force per unit mass: $g = F/m = GM/r^2$ (N kg⁻¹).

Near the Earth's surface — the field is (approximately) uniform and $g \approx 9.81$ N kg⁻¹ (height ≪ radius).

Effect of Earth's rotation — at the equator $mg = GMm/R^2 - mR\omega^2$ (the rotation term is tiny); at the poles $mg = GMm/R^2$.

Gravitational potential 引力势 φ — the work done per unit mass in bringing a mass from infinity to the point: $\varphi = -GM/r$ (J kg⁻¹). All values are negative (zero at infinity).

Gravitational potential energy 引力势能 — $GPE = m\varphi = -GMm/r$ (J). For a large distance use $\Delta GPE = m\varphi(\text{final}) - m\varphi(\text{initial})$; near the surface (g constant) $\Delta GPE = mgh$.

Field–potential relation — $g = -d\varphi/dr = -$(gradient of the φ–r graph); the minus sign shows φ decreases in the direction of g.

Circular orbit 圆轨道 — the centripetal force is provided by the gravitational force: $GMm/r^2 = mv^2/r \rightarrow v = \sqrt{(GM/r)}$.

First cosmic speed 第一宇宙速度 — the minimum orbital speed at the Earth's surface (r = R), $v \approx 7.9$ km s⁻¹.

Kepler's third law — r^3/T^2 = constant (∝ the central mass M).

Total energy of a satellite — $E = GPE + KE = -GMm/r + \frac{1}{2}mv^2$ (constant for a given orbit).

Escape speed 第二宇宙速度 — $v_E = \sqrt{(2GM/r)}$: the minimum speed to escape to infinity (KE + GPE = 0).

Geostationary satellite 同步卫星 — equatorial orbit, moves west → east with period T = 24 h, so it stays over the same point of the Earth.

点评 · Insight

天体圆轨道问题的统一思路是 "the centripetal force is provided by the gravitational force"：列 $GMm/r^2 = mr\omega^2 = mv^2/r$ 求解。$r^3/T^2 \propto M$ 说明轨道中心天体质量正比于 r^3/T^2。